

Sid Mathur's Thematic ETF Screener – Daily Report

2026-07-31

Contents

Action Center	2
Full Sample PC1	3
Position Sheet	4
Cluster Overview	5
Cluster Details	6
Long ETF Positions (full)	58
ETF Net Signal Strength	62
Diversifier Candidates	64
Run Parameters	65

Action Center**Run date:** 2026-07-31 **Universe:** 659 ETFs / 26 clusters**Full PC1:** FLAT **Clusters long:** 6/26 **ETFs long:** 168/659 (25%)**Today's events (cluster level)**

spread_buy = 5dma – 50dma; spread_sell = 5dma – 20dma (price units). A BUY fired today iff spread_buy crossed above 0; a SELL iff spread_sell crossed below the cushioned threshold.

BUY (1)

cluster	theme	spread_buy	p_h5	nss
cluster_4	Developed ex-US Equity	0.118	0.65	0.499

Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p_h5 is the 5-day probability for context. **spread_buy = 5dma – 50dma, spread_sell = 5dma – 20dma** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

Imminent BUY (top 5 by h=1 NSS, filter-approved)

cluster	theme	spread_buy	p_h1	p_h5	nss_h1	nss_h5
cluster_15	Real Estate	3.781	1.000	1.00	1.000	1.00
cluster_14	Financials	5.887	1.000	1.00	0.998	0.89
cluster_11	Healthcare	5.179	1.000	1.00	0.519	0.18
cluster_1	U.S. Mid Cap Equity	1.275	1.000	0.71	0.210	-0.17
cluster_12	Materials	-1.642	0.001	0.72	0.001	0.72

Full Sample PC1



Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.94 sticky

Current state: FLAT

5dma: 141.445 20dma: 142.030 50dma: 141.883

Spread vs 50dma: -0.437 Spread vs 20dma: -0.584

P(buy crossover within 5 days): 1.4% P(sell): 96.9%

Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	pos_wt	cor
VTV	cluster_3	U.S. Large Cap Value	1.000	75	7.8%	19.99%	0.98
BBAX	cluster_8	Asia-Pacific Equity	1.000	10	2.1%	9.34%	0.97
IGF	cluster_8	Asia-Pacific Equity	1.000	10	2.1%	5.57%	0.97
EWZ	cluster_8	Asia-Pacific Equity	1.000	10	2.1%	1.80%	0.97
PDBC	cluster_19	Broad Commodities	0.994	6	-2.5%	9.97%	0.98
XLP	cluster_18	Consumer Staples	0.944	23	-0.3%	10.66%	0.99
XLU	cluster_18	Consumer Staples	0.944	23	-0.3%	6.57%	0.99
ICSH	cluster_24	Cash / Ultra-Short Fixed In...	0.566	1	0.0%	8.05%	0.85
SHV	cluster_24	Cash / Ultra-Short Fixed In...	0.566	1	0.0%	5.51%	0.85
MINT	cluster_24	Cash / Ultra-Short Fixed In...	0.566	1	0.0%	3.45%	0.85
GBIL	cluster_24	Cash / Ultra-Short Fixed In...	0.566	1	0.0%	2.45%	0.85
VEA	cluster_4	Developed ex-US Equity	0.499	0	0.0%	13.99%	0.99

Watchlist (imminent-buy clusters, $p > 80\%$, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
XLF	cluster_14	Financials	1	0.889	1.095	0.96
VHT	cluster_11	Healthcare	1	0.183	1.089	0.93
VNQ	cluster_15	Real Estate	1	1.000	0.971	0.99
INDA	cluster_20	India Equity	1	0.569	0.660	0.91
SQQQ	cluster_26	Leveraged / Inverse	1	1.000	0.506	1.00
UUP	cluster_26	Leveraged / Inverse	1	1.000	0.495	1.00
NOBL	cluster_20	India Equity	1	0.569	0.080	0.91
PPLT	cluster_20	India Equity	1	0.569	0.029	0.91

Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
3	cluster_3	U.S. Large Cap Value	U.S. Large Cap Va...	68	LONG	shifted	75	7.8%	1.000	+1.297
8	cluster_8	Asia-Pacific Equity	Asia-Pacific Equi...	24	LONG	shifted	10	2.1%	1.000	+2.000
15	cluster_15	Real Estate	Real Estate	14	FLAT				1.000	+0.000
26	cluster_26	Leveraged / Inverse	Leveraged / Inver...	2	FLAT				1.000	+0.000
19	cluster_19	Broad Commodities	Broad Commodities	11	LONG		6	-2.5%	0.994	-0.006
18	cluster_18	Consumer Staples	Consumer Staples	11	LONG		23	-0.3%	0.944	+0.051
14	cluster_14	Financials	Financials	14	FLAT	shifted			0.889	+0.762
12	cluster_12	Materials	Materials	17	FLAT	shifted			0.723	+1.690
20	cluster_20	India Equity	India Equity	10	FLAT				0.569	+0.695
24	cluster_24	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	9	LONG		1	0.0%	0.566	+0.521
13	cluster_13	China Equity	China Equity	15	FLAT	shifted			0.506	+0.403
4	cluster_4	Developed ex-US Equity	Developed ex-US E...	61	LONG	shifted	0	0.0%	0.499	+1.497
11	cluster_11	Healthcare	Healthcare	17	FLAT	shifted			0.183	+1.183
1	cluster_1	U.S. Mid Cap Equity	U.S. Mid Cap Equi...	93	FLAT	shifted			-0.167	+0.832
17	cluster_17	Gold	Gold	12	FLAT				-0.253	+0.038
25	cluster_25	International Fixed Income	International Fix...	6	FLAT				-0.607	+0.358
21	cluster_21	Inflation-Protected Fixed I...	Inflation-Protect...	10	FLAT				-0.918	+0.045
10	cluster_10	Technology	Technology	18	FLAT	shifted			-0.944	-1.478
22	cluster_22	Japan Equity	Japan Equity	9	FLAT				-0.953	+0.028
23	cluster_23	Bank Loan / Floating Rate	Bank Loan / Float...	9	FLAT				-0.965	+0.033
5	cluster_5	Aggregate Fixed Income	Aggregate Fixed I...	58	FLAT				-0.989	+0.010
2	cluster_2	U.S. Large Cap Equity	U.S. Large Cap Eq...	74	FLAT	shifted			-1.000	-2.000
6	cluster_6	U.S. Large Cap Growth	U.S. Large Cap Gr...	38	FLAT	shifted			-1.000	-1.665
7	cluster_7	Emerging Markets Equity	Emerging Markets ...	25	FLAT	shifted			-1.000	+0.000
9	cluster_9	High Yield Bond	High Yield Bond	21	FLAT	shifted			-1.000	-1.998
16	cluster_16	Municipal Bond	Municipal Bond	13	FLAT				-1.000	+0.000

Cluster Details

Cluster 1 – U.S. Mid Cap Equity

Members: 93 **Var explained:** 0.863 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +29.6%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +19.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Industrials	11	0.1142
U.S. Small Cap Equity	10	0.1131
Consumer Discretionary	11	0.1082
U.S. Mid Cap Equity	6	0.0684
U.S. Small Cap Value	6	0.0673
U.S. Small Cap Growth	6	0.0662
Materials	4	0.0396
U.S. Mid Cap Value	3	0.0341

U.S. Mid Cap Growth	3	0.0334
Utilities	3	0.0301
Leveraged / Inverse	2	0.0216
Dividend Equity	2	0.0216
U.S. Small Cap Low Volatility	2	0.0205
Unknown	1	0.0115
U.S. Small Cap Fundamental Equity	1	0.0115
U.S. Small Cap ESG Equity	1	0.0114
U.S. Mid/Small Cap Equity	1	0.0114
Mid Cap Value Equities	1	0.0114
U.S. Small/Mid Cap Fundamental Equity	1	0.0114
U.S. Small Cap Multifactor Equity	1	0.0113
U.S. ESG Equity	1	0.0112
U.S. Mid Cap Dividend Equity	1	0.0111
U.S. Small/Mid Cap Equity	1	0.0111
Small Cap Value Equities	1	0.0111
U.S. Extended Market Equity	1	0.0111
U.S. Mid Cap Quality Equity	1	0.0110
Growth Equity	1	0.0110
U.S. Equity	1	0.0109
Value Equity	1	0.0109
U.S. Small Cap Dividend Equity	1	0.0109
U.S. Small Cap Industrials	1	0.0108
U.S. Small Cap Momentum Equity	1	0.0107
U.S. Micro Cap Equity	1	0.0105
U.S. Small Cap Consumer Discretionary	1	0.0103
U.S. Dividend Equity	1	0.0102
U.S. Mid Cap Low Volatility	1	0.0099
Real Estate	1	0.0089

Replication basket – IJH (cor=0.99) basket lead at lag 1: -0.010

ticker	rep_wt	name	category	etf_signal
IJH	0.994	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
31	MDY	0.0115	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	FLAT
39	IVOO	0.0115	Vanguard S&P Mid-Cap 400 ETF		FLAT
46	IJH	0.0115	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT

47	SPMD	0.0115	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	FLAT
49	VB	0.0115	Vanguard Small-Cap ETF	U.S. Small Cap Equity	FLAT
65	FNDA	0.0115	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	FLAT
17	JHMM	0.0114	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	FLAT
24	FNX	0.0114	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	FLAT
27	NUSC	0.0114	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	FLAT
28	PRFZ	0.0114	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	FLAT
40	VIOO	0.0114	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	FLAT
43	IJR	0.0114	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT
48	SPSM	0.0114	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	FLAT
61	EZM	0.0114	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	FLAT
70	IJJ	0.0114	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
72	IVOV	0.0114	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
73	MDYV	0.0114	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	FLAT
77	VBR	0.0114	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	FLAT
7	IWM	0.0113	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	FLAT
23	SCHM	0.0113	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	FLAT
25	FYX	0.0113	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	FLAT
29	SMLF	0.0113	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	FLAT
32	IJT	0.0113	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	FLAT
34	VIOG	0.0113	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	FLAT
37	SLYG	0.0113	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	FLAT
41	VTWO	0.0113	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	FLAT
50	SCHA	0.0113	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
68	IWN	0.0113	iShares Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
85	RWK	0.0113	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	FLAT
62	PSC	0.0112	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
71	IJS	0.0112	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
76	VTWV	0.0112	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
78	ETHO	0.0112	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	FLAT
92	SDVY	0.0112	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	LONG
11	FXR	0.0111	First Trust Industrials AlphaDEX	Industrials	FLAT
33	IJK	0.0111	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	FLAT
35	IVOG	0.0111	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	FLAT
36	MDYG	0.0111	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	FLAT
45	VXF	0.0111	Vanguard Extended Market ETF	U.S. Extended Market Equ...	FLAT
59	DON	0.0111	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	FLAT
60	EES	0.0111	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	FLAT
74	SLYV	0.0111	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	FLAT
75	VIOV	0.0111	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	FLAT
79	RWJ	0.0111	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	FLAT
93	ISMD	0.0111	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
44	ISCG	0.0110	iShares Morningstar Small-Cap Growth ETF	Growth Equity	FLAT
82	XMHQ	0.0110	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	FLAT
58	DES	0.0109	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	FLAT

84	RFV	0.0109	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	FLAT
86	OMFS	0.0109	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	FLAT
91	FYT	0.0109	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	FLAT
15	IYJ	0.0108	iShares U.S. Industrials ETF	Industrials	FLAT
21	VIS	0.0108	Vanguard Industrials ETF	Industrials	LONG
30	IWO	0.0108	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	FLAT
38	VTWG	0.0108	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	FLAT
51	FXD	0.0108	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	FLAT
80	PSCI	0.0108	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	FLAT
22	FIDU	0.0107	Fidelity MSCI Industrials Index ETF	Industrials	LONG
42	VBK	0.0107	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	FLAT
81	PAVE	0.0107	Global X U.S. Infrastructure Developmen...	Industrials	LONG
87	RZV	0.0107	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	FLAT
90	XSMO	0.0107	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	FLAT
56	RSPD	0.0106	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	LONG
26	IWC	0.0105	iShares Microcap ETF	U.S. Micro Cap Equity	FLAT
53	FTXR	0.0105	First Trust Nasdaq Transportation ETF	Industrials	LONG
69	SMMV	0.0105	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
2	XLI	0.0104	State Street Industrial Select Sector S...	Industrials	FLAT
14	FIW	0.0104	First Trust Water ETF	Utilities	FLAT
18	REGL	0.0104	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	LONG
89	PSCD	0.0103	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	FLAT
12	PHO	0.0102	Invesco Water Resources ETF	Utilities	FLAT
20	VAW	0.0102	Vanguard Materials ETF	Materials	FLAT
57	SMDV	0.0102	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT
64	XTN	0.0102	State Street SPDR S&P Transportation ETF	Industrials	FLAT
83	PKB	0.0102	Invesco Building & Construction ETF	Industrials	FLAT
67	XSLV	0.0101	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	FLAT
4	XRT	0.0100	State Street SPDR S&P Retail ETF	Consumer Discretionary	LONG
8	IYC	0.0100	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	FLAT
3	XLB	0.0099	State Street Materials Select Sector SP...	Materials	LONG
9	VCR	0.0099	Vanguard Consumer Discretion ETF	Consumer Discretionary	FLAT
10	FDIS	0.0099	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	FLAT
19	RSPM	0.0099	Invesco S&P 500 Equal Weight Materials ...	Materials	LONG
55	IYT	0.0099	iShares U.S. Transportation ETF	Industrials	FLAT
66	XMLV	0.0099	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	LONG
52	PEJ	0.0098	Invesco Leisure and Entertainment ETF	Consumer Discretionary	FLAT
63	XHB	0.0098	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	FLAT
16	IYM	0.0096	iShares U.S. Basic Materials ETF	Materials	FLAT
1	XLY	0.0095	State Street Consumer Discretionary Sel...	Consumer Discretionary	FLAT
13	CGW	0.0095	Invesco S&P Global Water Index ETF	Utilities	FLAT
5	JETS	0.0090	U.S. Global Jets ETF	Industrials	FLAT
54	ITB	0.0090	iShares U.S. Home Construction ETF	Consumer Discretionary	LONG
6	REM	0.0089	iShares Mortgage Real Estate ETF	Real Estate	FLAT
88	RTH	0.0089	VanEck Retail ETF	Consumer Discretionary	LONG

Cluster 2 – U.S. Large Cap Equity**Members:** 74 **Var explained:** 0.871 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +29.4%)
- Latest sell_cross: 2026-07-17
- Current 252d return: +16.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-17 more recent than last effective buy 2026-04-13)

**Category breakdown**

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1291
U.S. Total Market Equity	7	0.1008
Global Equity	7	0.0989
Technology	5	0.0654
Asset Allocation	4	0.0525
Industrials	4	0.0447
U.S. Equity	3	0.0421
U.S. Multifactor Equity	3	0.0420
U.S. Mega Cap Equity	3	0.0419
U.S. Quality / Factor Equity	3	0.0416
U.S. Mid Cap Growth	3	0.0414

U.S. Momentum Equity	3	0.0377
U.S. ESG Equity	2	0.0287
Momentum Equity	2	0.0261
Convertible Bond	2	0.0248
Commodity	1	0.0143
Large Cap Blend Equities	1	0.0142
Large Cap Value Equities	1	0.0142
Dividend Equity	1	0.0139
U.S. Equal Weight Equity	1	0.0137
U.S. Dividend Equity	1	0.0136
Growth Equity	1	0.0134
IPO Equity	1	0.0129
U.S. Moat Equity	1	0.0128
U.S. Mid Cap Momentum Equity	1	0.0128
Alternatives	1	0.0126
U.S. Small Cap Momentum Equity	1	0.0123
International IPO Equity	1	0.0111
Israel Equity	1	0.0106

Replication basket – VOO (cor=0.98) basket lead at lag 1: -0.011

ticker	rep_wt	name	category	etf_signal
VOO	0.995	Vanguard S&P 500 ETF	U.S. Large Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
44	IWV	0.0144	iShares Russell 3000 Fund	U.S. Total Market Equity	FLAT
45	IYY	0.0144	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	FLAT
50	ESGU	0.0144	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	FLAT
51	IWB	0.0144	iShares Russell 1000 ETF	U.S. Large Cap Equity	FLAT
56	VTHR	0.0144	Vanguard Russell 3000 ETF	U.S. Total Market Equity	FLAT
59	VONE	0.0144	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	FLAT
62	ILCB	0.0144	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	FLAT
63	ITOT	0.0144	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	FLAT
65	SCHB	0.0144	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	FLAT
66	SCHX	0.0144	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	FLAT
67	SPTM	0.0144	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	FLAT
69	VTI	0.0144	Vanguard Total Stock Market ETF	U.S. Total Market Equity	FLAT
3	SPY	0.0143	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	FLAT
5	ACWI	0.0143	iShares MSCI ACWI ETF	Global Equity	FLAT

10	URTH	0.0143	iShares MSCI World ETF	Global Equity	FLAT
13	VT	0.0143	Vanguard Total World Stock Index ETF	Global Equity	FLAT
31	TILT	0.0143	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	FLAT
40	JHML	0.0143	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	FLAT
43	SUSA	0.0143	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	FLAT
46	LRGF	0.0143	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	FLAT
48	SPYX	0.0143	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	FLAT
57	GSLC	0.0143	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	FLAT
61	VV	0.0143	Vanguard Large-Cap ETF	U.S. Large Cap Equity	FLAT
64	IVV	0.0143	iShares Core S&P 500 ETF	U.S. Large Cap Equity	FLAT
68	VOO	0.0143	Vanguard S&P 500 ETF	U.S. Large Cap Equity	FLAT
11	CRBN	0.0142	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	FLAT
12	SPGM	0.0142	State Street SPDR Portfolio MSCI Global...	Global Equity	FLAT
42	DSI	0.0142	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	FLAT
52	IWL	0.0142	iShares Russell Top 200 ETF	U.S. Equity	FLAT
58	EPS	0.0142	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	FLAT
60	MGC	0.0142	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	FLAT
6	AOA	0.0141	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	FLAT
47	OEF	0.0140	iShares S&P 100 Fund	U.S. Mega Cap Equity	FLAT
53	QUAL	0.0140	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	FLAT
29	IMCG	0.0139	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	FLAT
38	QDF	0.0139	FlexShares Quality Dividend Index Fund	Dividend Equity	FLAT
28	VOT	0.0138	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	FLAT
39	CATH	0.0138	Global X S&P 500 Catholic Values ETF	Global Equity	FLAT
54	QUS	0.0138	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	FLAT
55	SPHQ	0.0138	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	FLAT
9	AOR	0.0137	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	FLAT
16	QQEW	0.0137	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	FLAT
17	XT	0.0137	iShares Future Exponential Technologies...	Technology	FLAT
25	IWP	0.0137	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	FLAT
37	IOO	0.0137	iShares Global 100 ETF	Global Equity	FLAT
49	XLG	0.0137	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	FLAT
19	FTC	0.0136	First Trust Large Cap Growth AlphaDEX F...	Technology	FLAT
36	SPHB	0.0136	Invesco S&P 500 High Beta ETF	U.S. Equity	FLAT
41	DGRW	0.0136	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	FLAT
74	OMFL	0.0135	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	FLAT
23	RPG	0.0134	Invesco S&P 500 Pure Growth ETF	Growth Equity	FLAT
2	ROBO	0.0133	ROBO Global Robotics and Automation Ind...	Technology	FLAT
20	PDP	0.0132	Invesco Dorsey Wright Momentum ETF	Momentum Equity	FLAT
71	VFMO	0.0131	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	FLAT
1	BOTZ	0.0130	Global X Robotics & Artificial Intellig...	Technology	FLAT
21	FPX	0.0129	First Trust US Equity Opportunities ETF	IPO Equity	FLAT
8	AOM	0.0128	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	FLAT
14	FV	0.0128	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	FLAT
24	XMMO	0.0128	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	FLAT

30	MOAT	0.0128	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	FLAT
27	MTUM	0.0127	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	FLAT
4	QAI	0.0126	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	FLAT
22	CWB	0.0126	State Street SPDR Bloomberg Convertible...	Convertible Bond	FLAT
32	DWAS	0.0123	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	FLAT
26	ICVT	0.0121	iShares Convertible Bond ETF	Convertible Bond	FLAT
73	PRN	0.0120	Invesco Dorsey Wright Industrials Momen...	Industrials	FLAT
15	PRNT	0.0119	3D Printing (The) ETF	Technology	FLAT
70	QMOM	0.0119	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	FLAT
7	AOK	0.0118	iShares Core 30/70 Conservative Allocat...	Asset Allocation	FLAT
35	XAR	0.0112	State Street SPDR S&P Aerospace & Defen...	Industrials	FLAT
18	FPXI	0.0111	First Trust International Equity Opport...	International IPO Equity	FLAT
33	PPA	0.0111	Invesco Aerospace & Defense ETF	Industrials	LONG
72	EIS	0.0106	iShares MSCI Israel ETF	Israel Equity	FLAT
34	ITA	0.0104	iShares U.S. Aerospace & Defense ETF	Industrials	LONG

Cluster 3 – U.S. Large Cap Value

Members: 68 **Var explained:** 0.886 **Signal:** LONG since 2026-04-13 (75 days, 7.8%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +23.1%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +20.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
U.S. Dividend Equity	21	0.3063
U.S. Large Cap Value	10	0.1536
Value Equity	4	0.0594
U.S. Low Volatility Equity	4	0.0547
U.S. Multifactor Equity	3	0.0450
U.S. Mid Cap Equity	3	0.0441
U.S. Value Factor Equity	3	0.0437
U.S. Fundamental Equity	2	0.0306
U.S. Mid Cap Value	2	0.0305
U.S. Equal Weight Equity	2	0.0303
U.S. Equity	2	0.0303
Dividend Equity	2	0.0282
U.S. Mega Cap Value	1	0.0152
U.S. Revenue Weighted Equity	1	0.0151
Large Cap Blend Equities	1	0.0149
U.S. Size Factor Equity	1	0.0149
U.S. Buyback Equity	1	0.0147
U.S. Large Cap Equity	1	0.0146
U.S. Quality / Factor Equity	1	0.0143
U.S. Defensive Equity	1	0.0140
Global Equity	1	0.0135

Communication Services 1 0.0121

Replication basket – VTV (cor=0.98) basket lead at lag 1: 0.006

ticker	rep_wt	name	category	etf_signal
VTV	1	Vanguard Value ETF	U.S. Large Cap Value	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
24	IWD	0.0154	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
27	JPUS	0.0154	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	FLAT
32	VONV	0.0154	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
38	IUSV	0.0154	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	FLAT
39	SCHV	0.0154	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	FLAT
42	VTV	0.0154	Vanguard Value ETF	U.S. Large Cap Value	LONG
59	DTD	0.0154	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
2	RSP	0.0153	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
9	PRF	0.0153	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
14	CFA	0.0153	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	FLAT
16	NULV	0.0153	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	FLAT
20	FNDX	0.0153	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
26	IVE	0.0153	iShares S&P 500 Value ETF	U.S. Large Cap Value	FLAT
30	VOOV	0.0153	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	FLAT
35	VYM	0.0153	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	FLAT
36	ILCV	0.0153	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	FLAT
40	SPYV	0.0153	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	FLAT
54	VOE	0.0153	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
64	DGRO	0.0153	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
22	IWX	0.0152	iShares Russell Top 200 Value ETF	Value Equity	LONG
33	MGV	0.0152	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	FLAT
51	IWS	0.0152	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
58	DLN	0.0152	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
10	RWL	0.0151	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
12	AIVL	0.0151	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
23	ONEV	0.0151	State Street SPDR Russell 1000 Low Vola...	U.S. Equity	FLAT
43	FTA	0.0151	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	LONG
56	PFM	0.0151	Invesco Dividend Achievers ETF	U.S. Dividend Equity	FLAT
67	JPME	0.0151	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	FLAT
21	EQAL	0.0150	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	FLAT
52	ONEY	0.0150	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	FLAT

5	FEX	0.0149	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
18	FDVV	0.0149	Fidelity High Dividend ETF	U.S. Dividend Equity	FLAT
28	SIZE	0.0149	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	FLAT
65	VIG	0.0149	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
17	SDY	0.0148	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
34	SCHD	0.0148	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
37	IMCB	0.0148	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
4	PKW	0.0147	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG
6	PWV	0.0147	Invesco Large Cap Value ETF	Value Equity	LONG
25	IWR	0.0147	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
41	VO	0.0147	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
46	SDOG	0.0147	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	FLAT
47	DVY	0.0147	iShares Select Dividend ETF	U.S. Dividend Equity	FLAT
68	JVAL	0.0147	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
3	FVD	0.0146	First Trust VL Dividend	U.S. Dividend Equity	LONG
61	DIA	0.0146	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	FLAT
7	RDVY	0.0145	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	FLAT
49	RPV	0.0144	Invesco S&P 500 Pure Value ETF	Value Equity	FLAT
53	SPYD	0.0144	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	FLAT
57	OUSA	0.0144	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
63	LGLV	0.0144	State Street SPDR US Large Cap Low Vola...	U.S. Low Volatility Equi...	LONG
15	CFO	0.0143	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	FLAT
55	FTCS	0.0143	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	LONG
11	DHS	0.0142	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	FLAT
62	USMV	0.0141	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	FLAT
13	CDC	0.0140	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	FLAT
48	RDIV	0.0140	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	FLAT
29	VLUE	0.0139	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
50	SPHD	0.0138	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
8	FDL	0.0137	First Trust Morningstar ETF	U.S. Dividend Equity	LONG
44	PEY	0.0137	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	FLAT
1	ACWV	0.0135	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG
45	DIV	0.0135	Global X Super Dividend ETF	U.S. Dividend Equity	LONG
31	HDV	0.0134	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
19	LVHD	0.0133	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	LONG
60	SPLV	0.0129	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	LONG
66	IYZ	0.0121	iShares U.S. Telecommunications ETF	Communication Services	FLAT

Cluster 4 – Developed ex-US Equity

Members: 61 Var explained: 0.894 Signal: LONG since 2026-07-30 (0 days, 0.0%)

Signal logic:

- Latest buy_cross: 2026-07-30 – filter **PASS** (252d at date = +21.9%)
- Latest sell_cross: 2026-06-25
- Current 252d return: +21.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-30 > last sell at 2026-06-25)

**Category breakdown**

category	n_members	abs_weight
International Equity	19	0.3157
Developed ex-US Equity	9	0.1528
Global ex-US Equity	5	0.0848
Europe Equity	4	0.0675
International Dividend Equity	4	0.0659
Alternatives	4	0.0616
Eurozone Equity	3	0.0490
Commodity	1	0.0172
Global ex-US Small Cap Equity	1	0.0166
Germany Equity	1	0.0162
France Equity	1	0.0162
Italy Equity	1	0.0159
Europe Financials	1	0.0158
United Kingdom Equity	1	0.0157
Netherlands Equity	1	0.0156

Sweden Equity	1	0.0156
Spain Equity	1	0.0151
Austria Equity	1	0.0148
Belgium Equity	1	0.0145
Poland Equity	1	0.0135

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.020

ticker	rep_wt	name	category	etf_signal
VEA	0.959	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	EFA	0.0173	iShares MSCI EAFE ETF	Developed ex-US Equity	LONG
51	IEFA	0.0173	iShares Core MSCI EAFE ETF	Developed ex-US Equity	LONG
3	VEA	0.0172	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT
40	GSIE	0.0172	Goldman Sachs ActiveBeta International ...	Commodity	LONG
41	ESGD	0.0172	iShares ESG Aware MSCI EAFE ETF	International Equity	LONG
52	SCHF	0.0172	Schwab International Equity ETF	Developed ex-US Equity	FLAT
53	SPDW	0.0172	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	FLAT
25	TLTD	0.0171	FlexShares Morningstar Developed Market...	International Equity	LONG
32	INTF	0.0171	iShares International Equity Factor ETF	International Equity	LONG
6	IEV	0.0170	iShares Europe ETF	Europe Equity	LONG
16	DWM	0.0170	WisdomTree International Equity Fund	International Dividend E...	LONG
19	PXF	0.0170	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	FLAT
29	ACWX	0.0170	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	FLAT
46	IEUR	0.0170	iShares Core MSCI Europe ETF	Europe Equity	LONG
47	IXUS	0.0170	iShares Core MSCI Total International S...	Global ex-US Equity	FLAT
48	VEU	0.0170	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	FLAT
49	VGK	0.0170	Vanguard FTSEEurope ETF	Europe Equity	LONG
50	VXUS	0.0170	Vanguard Total International Stock ETF	Global ex-US Equity	FLAT
30	CWI	0.0169	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	FLAT
33	IQLT	0.0169	iShares MSCI Intl Quality Factor ETF	International Equity	LONG
35	QEFA	0.0169	State Street SPDR MSCI EAFE StrategicFa...	International Equity	LONG
39	FNDF	0.0169	Schwab Fundamental International Equity...	International Equity	FLAT
14	EZU	0.0168	iShares MSCI Eurozone ETF	Eurozone Equity	FLAT
23	EFV	0.0168	iShares MSCI EAFE Value ETF	Developed ex-US Equity	LONG
26	JPIN	0.0168	JPMorgan Diversified Return Internation...	International Equity	LONG
38	VYMI	0.0168	Vanguard International High Dividend Yi...	International Dividend E...	LONG
18	IQDF	0.0167	FlexShares International Quality Divide...	International Equity	LONG

22	SCZ	0.0167	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	FLAT
24	FNDC	0.0167	Schwab Fundamental International Small ...	International Equity	FLAT
34	IVLU	0.0167	iShares MSCI Intl Value Factor ETF	International Equity	LONG
37	RODM	0.0167	Hartford Multifactor Developed Markets ...	International Equity	LONG
44	SCHC	0.0167	Schwab International Small-Cap Equity E...	Developed ex-US Equity	FLAT
56	HFXI	0.0167	NYLI FTSE International Equity Currency...	International Equity	FLAT
20	EFG	0.0166	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	FLAT
36	FEZ	0.0166	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	LONG
45	VSS	0.0166	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	FLAT
4	FEP	0.0165	First Trust Europe AlphaDEX Fund	Europe Equity	FLAT
7	DLS	0.0165	WisdomTree International SmallCap Fund	International Equity	FLAT
31	IMTM	0.0163	iShares MSCI Intl Momentum Factor ETF	International Equity	FLAT
43	VIGI	0.0163	Vanguard International Dividend Appreci...	International Dividend E...	LONG
1	EWG	0.0162	iShares MSCI Germany Index Fund	Germany Equity	FLAT
12	EWQ	0.0162	iShares MSCI France Index Fund	France Equity	LONG
21	GWX	0.0161	State Street SPDR S&P International Sma...	International Equity	FLAT
5	VIDI	0.0160	Vident International Equity Strategy ETF	International Equity	FLAT
59	IPKW	0.0160	Invesco International BuyBack Achievers...	International Equity	LONG
61	EWI	0.0159	iShares MSCI Italy ETF	Italy Equity	FLAT
15	IDV	0.0158	iShares International Select Dividend E...	International Dividend E...	LONG
17	EUFN	0.0158	iShares MSCI Europe Financials ETF	Europe Financials	LONG
13	EWU	0.0157	iShares MSCI United Kingdom ETF	United Kingdom Equity	LONG
27	DBEF	0.0157	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
10	HEZU	0.0156	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	LONG
28	HEFA	0.0156	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
42	HDEF	0.0156	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	LONG
55	EWD	0.0156	iShares MSCI Sweden ETF	Sweden Equity	LONG
58	EWN	0.0156	iShares MSCI Netherlands Index Fund	Netherlands Equity	FLAT
8	HEDJ	0.0151	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
9	IHDG	0.0151	WisdomTree International Hedged Quality...	Alternatives	FLAT
11	EWP	0.0151	iShares MSCI Spain ETF	Spain Equity	FLAT
57	EWO	0.0148	iShares MSCI Austria ETF	Austria Equity	FLAT
60	EWK	0.0145	iShares MSCI Belgium ETF	Belgium Equity	LONG
54	EPOL	0.0135	iShares MSCI Poland ETF	Poland Equity	LONG

Cluster 5 – Aggregate Fixed Income

Members: 58 **Var explained:** 0.831 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +4.5%)

- Latest sell_cross: 2026-07-07
- Current 252d return: +1.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)



Category breakdown

category	n_members	abs_weight
Fixed Income	19	0.3284
Investment Grade Corporate Bond	13	0.2303
Aggregate Fixed Income	6	0.1120
Mortgage-Backed Fixed Income	6	0.0982
Active Fixed Income	5	0.0891
Cash / Ultra-Short Fixed Income	3	0.0470
International Fixed Income	2	0.0313
Target Maturity Bond	2	0.0306
Agency Fixed Income	1	0.0166
Taxable Municipal Bond	1	0.0165

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.007

ticker	rep_wt	name	category	etf_signal
BND	1.007	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	BIV	0.0188	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
26	AGG	0.0188	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	FLAT
28	BND	0.0188	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT
30	SCHZ	0.0188	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	FLAT
20	IUSB	0.0187	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	FLAT
31	SPAB	0.0187	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	FLAT
2	IEF	0.0185	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
17	IGIB	0.0184	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	FLAT
24	VCIT	0.0184	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	FLAT
7	FBND	0.0183	Fidelity Total Bond ETF	Active Fixed Income	FLAT
43	SPTI	0.0183	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
46	SCHR	0.0183	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
47	VGIT	0.0183	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
13	AGGY	0.0182	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	FLAT
15	SPIB	0.0182	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	FLAT
6	GTO	0.0181	Invesco Total Return Bond ETF	Active Fixed Income	FLAT
11	GVI	0.0181	iShares Intermediate Government/Credit ...	Fixed Income	FLAT
40	IEI	0.0181	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
50	MBB	0.0181	iShares MBS ETF	Mortgage-Backed Fixed In...	FLAT
12	QLTA	0.0180	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	FLAT
27	USIG	0.0180	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	FLAT
51	VMBS	0.0180	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	FLAT
52	SPMB	0.0180	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	FLAT
3	LQD	0.0179	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	FLAT
5	BOND	0.0179	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	FLAT
19	ISTB	0.0178	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT
8	FLCO	0.0177	Franklin Investment Grade Corporate ETF	Active Fixed Income	FLAT
10	CORP	0.0177	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	FLAT
22	BLV	0.0176	Vanguard Long-Term Bond ETF	Fixed Income	FLAT
41	TLH	0.0176	iShares 10-20 Year Treasury Bond ETF	Fixed Income	FLAT
45	GOVT	0.0176	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
35	IGSB	0.0175	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
37	VCSH	0.0175	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
23	BSV	0.0174	Vanguard Short-Term Bond ETF	Fixed Income	FLAT
49	GNMA	0.0174	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	FLAT
18	IGLB	0.0172	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	FLAT
44	SPTL	0.0172	State Street SPDR Portfolio Long Term T...	Fixed Income	FLAT
16	SPLB	0.0171	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	FLAT
25	VCLT	0.0171	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
36	SLQD	0.0171	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
48	VGLT	0.0171	Vanguard Long-Term Treasury ETF	Fixed Income	FLAT

29	TOTL	0.0170	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
1	TLT	0.0169	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT
39	AGZ	0.0166	iShares Agency Bond ETF	Agency Fixed Income	FLAT
9	BAB	0.0165	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	FLAT
42	EDV	0.0163	Vanguard Extended Duration Treasury ETF	Fixed Income	FLAT
34	SPSB	0.0162	State Street SPDR Portfolio Short Term ...	Fixed Income	FLAT
53	SHY	0.0162	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
56	VGSH	0.0159	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
57	BNDX	0.0158	Vanguard Total International Bond ETF	International Fixed Inco...	FLAT
55	SCHO	0.0156	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
58	IAGG	0.0156	iShares International Aggregate Bond Fu...	International Fixed Inco...	FLAT
33	BSCQ	0.0155	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	FLAT
54	SPTS	0.0155	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
14	IBDR	0.0151	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	LONG
38	CMBS	0.0135	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	FLAT
4	LMBS	0.0133	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	FLAT
32	LDUR	0.0121	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	FLAT

Cluster 6 – U.S. Large Cap Growth

Members: 38 **Var explained:** 0.894 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +56.4%)
- Latest sell_cross: 2026-07-16
- Current 252d return: +28.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-16 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Technology	19	0.4990
U.S. Large Cap Growth	11	0.2989
Growth Equity	2	0.0536
U.S. Equity	1	0.0273
U.S. Mega Cap Growth	1	0.0270
Options Income	1	0.0246
U.S. Small Cap Technology	1	0.0243
Trend-Following Equity	1	0.0242
U.S. Trend-Following Equity	1	0.0211

Replication basket – QQQ (cor=0.98) basket lead at lag 1: -0.032

ticker	rep_wt	name	category	etf_signal
QQQ	1.032	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	QQQ	0.0275	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	FLAT
16	IGM	0.0275	iShares Expanded Tech Sector ETF	Technology	FLAT
18	IYW	0.0275	iShares U.S. Technology ETF	Technology	FLAT
24	VGT	0.0275	Vanguard Information Tech ETF	Technology	FLAT
26	FTEC	0.0275	Fidelity MSCI Information Technology In...	Technology	FLAT
29	ILCG	0.0274	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
1	XLK	0.0273	State Street Technology Select Sector S...	Technology	FLAT
17	IXN	0.0273	iShares Global Tech ETF	Technology	FLAT
20	ONEQ	0.0273	Fidelity Nasdaq Composite Index ETF	U.S. Equity	FLAT
23	IVW	0.0272	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	FLAT
25	VOOG	0.0272	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	FLAT
27	VONG	0.0272	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	FLAT
30	IUSG	0.0272	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	FLAT
32	SPYG	0.0272	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	FLAT
12	XNTK	0.0271	State Street SPDR NYSE Technology ETF	Technology	FLAT
22	IWF	0.0271	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	FLAT
33	VUG	0.0271	Vanguard Growth ETF	U.S. Large Cap Growth	FLAT
19	NULG	0.0270	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
21	IWY	0.0270	iShares Russell Top 200 Growth ETF	Growth Equity	FLAT
28	MGK	0.0270	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	FLAT
7	QTEC	0.0269	First Trust NASDAQ-100-Technology Secto...	Technology	FLAT
31	SCHG	0.0269	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
4	AIQ	0.0266	Global X Artificial Intelligence & Tech...	Technology	FLAT
11	FXL	0.0266	First Trust Technology AlphaDEX	Technology	FLAT
15	PWB	0.0266	Invesco Large Cap Growth ETF	Growth Equity	FLAT
3	SMH	0.0259	VanEck Semiconductor ETF	Technology	FLAT
9	TDIV	0.0259	First Trust NASDAQ Technology Dividend ...	Technology	FLAT
2	SOXX	0.0256	iShares PHLX SOX Semiconductor Sector I...	Technology	FLAT
38	QTUM	0.0255	Defiance Quantum ETF	Technology	LONG
10	XSD	0.0250	State Street SPDR S&P Semiconductor ETF	Technology	FLAT
36	FTXL	0.0250	First Trust Nasdaq Semiconductor ETF	Technology	FLAT
8	PSI	0.0249	Invesco Semiconductors ETF	Technology	FLAT
6	NXTG	0.0248	First Trust Indxx NextG ETF	Technology	FLAT
14	QYLD	0.0246	Global X NASDAQ 100 Covered Call ETF	Options Income	FLAT
37	PTF	0.0244	Invesco Dorsey Wright Technology Moment...	Technology	LONG
35	PSCT	0.0243	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	FLAT
13	PTNQ	0.0242	Pacer Trendpilot 100 ETF	Trend-Following Equity	FLAT
34	PTLC	0.0211	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	FLAT

Cluster 7 – Emerging Markets Equity**Members:** 25 **Var explained:** 0.871 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +57.0%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +24.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-04-10)

**Category breakdown**

category	n_members	abs_weight
Emerging Markets Equity	19	0.7729
Asia Equity	2	0.0815
Asia ex-Japan Equity	1	0.0418
Taiwan Equity	1	0.0362
South Korea Equity	1	0.0340
Global Dividend Equity	1	0.0336

Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.036

ticker	rep_wt	name	category	etf_signal
IEMG	0.918	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	IEMG	0.0425	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	FLAT
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	FLAT
20	ESGE	0.0423	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	FLAT
23	SPEM	0.0423	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	FLAT
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	FLAT
16	GEM	0.0422	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	FLAT
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	FLAT
18	XSOE	0.0420	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	FLAT
15	EMGF	0.0419	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	FLAT
5	AAXJ	0.0418	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	FLAT
11	EEMA	0.0415	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	FLAT
14	GMF	0.0412	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	FLAT
12	PXH	0.0409	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	FLAT
17	FNDE	0.0409	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	LONG
19	EEMV	0.0405	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	FLAT
8	DGS	0.0403	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	FLAT
10	AIA	0.0403	iShares Asia 50 ETF	Asia Equity	LONG
7	DEM	0.0397	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	FLAT
6	EWX	0.0392	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	FLAT
4	FEM	0.0389	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	FLAT
13	DVYE	0.0364	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	LONG
9	EWI	0.0362	iShares MSCI Taiwan ETF	Taiwan Equity	FLAT
25	FEMS	0.0349	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	FLAT
1	EWY	0.0340	iShares MSCI South Korea ETF	South Korea Equity	FLAT
24	SDIV	0.0336	Global X SuperDividend ETF	Global Dividend Equity	LONG

Cluster 8 – Asia-Pacific Equity

Members: 24 **Var explained:** 0.680 **Signal:** LONG since 2026-07-16 (10 days, 2.1%)

Signal logic:

- Latest buy_cross: 2026-07-16 – filter **PASS** (252d at date = +14.5%)

- Latest sell_cross: 2026-06-23
- Current 252d return: +15.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-16 > last sell at 2026-06-23)



Category breakdown

category	n_members	abs_weight
Asia-Pacific Equity	2	0.0945
International Equity	2	0.0919
International Dividend Equity	2	0.0918
Real Estate	2	0.0881
Global Equity	2	0.0880
Emerging Markets Fixed Income	2	0.0804
Currency	2	0.0777
Australia Equity	1	0.0467
Canada Equity	1	0.0444
Switzerland Equity	1	0.0414
Singapore Equity	1	0.0399
South Africa Equity	1	0.0395
Latin America Equity	1	0.0379
New Zealand Equity	1	0.0379
Mexico Equity	1	0.0359
Brazil Equity	1	0.0321
Chile Equity	1	0.0319

Replication basket – EWZ+IGF+BBAX (cor=0.97) basket lead at lag 1: -0.006

ticker	rep_wt	name	category	etf_signal
EWZ	0.090	iShares MSCI Brazil ETF	Brazil Equity	LONG
IGF	0.278	iShares Global Infrastructure ETF	Global Equity	LONG
BBAX	0.467	JPMorgan BetaBuilders Developed Asia ...	Asia-Pacific Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
8	EPP	0.0473	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	LONG
24	BBAX	0.0472	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	LONG
12	IDLV	0.0469	Invesco S&P International Developed Low...	International Equity	LONG
4	EWA	0.0467	iShares MSCI Australia Index Fund	Australia Equity	LONG
11	DWX	0.0463	State Street SPDR S&P International Div...	International Dividend E...	LONG
3	PID	0.0455	Invesco International Dividend Achiever...	International Dividend E...	LONG
13	EFAV	0.0450	iShares MSCI EAFE Min Vol Factor ETF	International Equity	LONG
9	NFRA	0.0446	FlexShares STOXX Global Broad Infrastru...	Global Equity	LONG
14	VNQI	0.0446	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	LONG
7	EWC	0.0444	iShares MSCI Canada Index Fund	Canada Equity	LONG
2	RWX	0.0435	State Street SPDR Dow Jones Internation...	Real Estate	LONG
10	IGF	0.0434	iShares Global Infrastructure ETF	Global Equity	LONG
5	EWL	0.0414	iShares MSCI Switzerland ETF	Switzerland Equity	LONG
20	FXA	0.0414	Invesco CurrencyShares Australian Dolla...	Currency	FLAT
15	EBND	0.0403	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	FLAT
16	EMLC	0.0401	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	FLAT
6	EWS	0.0399	iShares MSCI Singapore ETF	Singapore Equity	LONG
22	EZA	0.0395	iShares MSCI South Africa Index Fund	South Africa Equity	FLAT
19	ILF	0.0379	iShares Latin America 40 ETF	Latin America Equity	LONG
21	ENZL	0.0379	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
23	FXC	0.0363	Invesco CurrencyShares Canadian Dollar ...	Currency	FLAT
17	EWV	0.0359	iShares MSCI Mexico ETF	Mexico Equity	FLAT
1	EWZ	0.0321	iShares MSCI Brazil ETF	Brazil Equity	LONG
18	ECH	0.0319	iShares MSCI Chile ETF	Chile Equity	FLAT

Cluster 9 – High Yield Bond

Members: 21 **Var explained:** 0.765 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +7.3%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +3.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.5157
Preferred Stock	7	0.3058
Emerging Markets Fixed Income	4	0.1785

Replication basket – HYG (cor=0.95) basket lead at lag 1: 0.046

ticker	rep_wt	name	category	etf_signal
HYG	0.917	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
--------	-------	------	----------	------------

13	JNK	0.0527	State Street SPDR Bloomberg High Yield ...	High Yield Bond	FLAT
1	HYG	0.0526	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	FLAT
19	HYLB	0.0524	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	FLAT
18	BBHY	0.0523	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	FLAT
16	SHYG	0.0521	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	FLAT
14	SJNK	0.0520	State Street SPDR Bloomberg Short Term ...	High Yield Bond	FLAT
17	FALN	0.0517	iShares Fallen Angels USD Bond ETF	High Yield Bond	FLAT
15	ANGL	0.0516	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	FLAT
6	HYS	0.0511	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	FLAT
9	PFF	0.0479	iShares Preferred and Income Securities...	Preferred Stock	FLAT
3	HYLS	0.0472	First Trust Tactical High Yield ETF	High Yield Bond	FLAT
2	EMB	0.0468	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
21	VWOB	0.0467	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	FLAT
11	PFXF	0.0459	VanEck Preferred Securities ex Financia...	Preferred Stock	FLAT
20	PCY	0.0458	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	FLAT
7	PGX	0.0446	Invesco Preferred ETF	Preferred Stock	FLAT
10	PSK	0.0441	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0433	First Trust Preferred Securities and In...	Preferred Stock	FLAT
5	PGF	0.0431	Invesco Financial Preferred ETF	Preferred Stock	FLAT
12	HYEM	0.0393	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	FLAT
8	VRP	0.0368	Invesco Variable Rate Preferred ETF	Preferred Stock	FLAT

Cluster 10 – Technology

Members: 18 **Var explained:** 0.805 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-07-01 – filter **PASS** (252d at date = +1.6%)
- Latest sell_cross: 2026-07-23
- Current 252d return: -2.1% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-07-01)



Category breakdown

category	n_members	abs_weight
Technology	10	0.5586
Communication Services	3	0.1636
Financials	2	0.1126
Consumer Discretionary	2	0.1082
IPO Equity	1	0.0569

Replication basket – XLC+IGV (cor=0.94) basket lead at lag 1: 0.032

ticker	rep_wt	name	category	etf_signal
XLC	0.508	State Street Communication Services S...	Communication Services	FLAT
IGV	0.600	iShares Expanded Tech-Software Sector...	Technology	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
14	FDN	0.0597	First Trust DJ Internet Index Fund	Technology	FLAT
13	PNQI	0.0592	Invesco Nasdaq Internet ETF	Technology	FLAT

7	ARKW	0.0582	ARK Next Generation Internet ETF	Technology	FLAT
16	FINX	0.0580	Global X FinTech ETF	Financials	FLAT
3	SKYY	0.0578	First Trust Cloud Computing ETF	Technology	LONG
6	ARKK	0.0570	ARK Innovation ETF	Technology	FLAT
12	IPO	0.0569	Renaissance IPO ETF	IPO Equity	FLAT
10	IBUY	0.0567	Amplify Online Retail ETF	Consumer Discretionary	FLAT
2	IGV	0.0562	iShares Expanded Tech-Software Sector E...	Technology	FLAT
18	FCOM	0.0552	Fidelity MSCI Communication Services In...	Communication Services	FLAT
17	VOX	0.0551	Vanguard Communication Services ETF	Communication Services	FLAT
15	IPAY	0.0546	Amplify Digital Payments ETF	Financials	FLAT
8	ARKQ	0.0541	ARK Autonomous Technology & Robotics ETF	Technology	FLAT
11	CIBR	0.0536	First Trust NASDAQ Cybersecurity ETF	Technology	FLAT
1	XLC	0.0532	State Street Communication Services Sel...	Communication Services	FLAT
5	HACK	0.0528	Amplify Cybersecurity ETF	Technology	FLAT
9	ESPO	0.0516	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
4	BLOK	0.0499	Amplify Blockchain Technology ETF	Technology	FLAT

Cluster 11 – Healthcare

Members: 17 **Var explained:** 0.757 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.8%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +29.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9418
U.S. Small Cap Healthcare	1	0.0582

Replication basket – VHT (cor=0.93) basket lead at lag 1: -0.004

ticker	rep_wt	name	category	etf_signal
VHT	1.089	Vanguard Health Care ETF	Healthcare	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	VHT	0.0655	Vanguard Health Care ETF	Healthcare	LONG
16	FHLC	0.0655	Fidelity MSCI Health Care Index ETF	Healthcare	LONG
9	FXH	0.0648	First Trust Health Care AlphaDEX	Healthcare	FLAT
14	RSPH	0.0641	Invesco S&P 500 Equal Weight Health Car...	Healthcare	FLAT
13	IYH	0.0639	iShares U.S. Healthcare ETF	Healthcare	LONG
10	IXJ	0.0625	iShares Global Healthcare ETF	Healthcare	FLAT

1	XLV	0.0622	State Street Health Care Select Sector ...	Healthcare	LONG
3	IBB	0.0605	iShares Biotechnology ETF	Healthcare	FLAT
6	FBT	0.0597	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0582	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	FLAT
7	XHE	0.0574	State Street SPDR S&P Health Care Equip...	Healthcare	FLAT
12	IHI	0.0567	iShares U.S. Medical Devices ETF	Healthcare	FLAT
17	PPH	0.0545	VanEck Pharmaceutical ETF	Healthcare	LONG
5	PTH	0.0536	Invesco Dorsey Wright Healthcare Moment...	Healthcare	LONG
4	XBI	0.0531	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
2	ARKG	0.0505	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0473	iShares U.S. Health Care Providers ETF	Healthcare	LONG

Cluster 12 – Materials

Members: 17 **Var explained:** 0.686 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +49.0%)
- Latest sell_cross: 2026-06-09
- Current 252d return: +29.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-09 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Materials	9	0.5592
MLP / Energy Infrastructure	4	0.2177
Energy	1	0.0644
Consumer Staples	1	0.0597
Norway Equity	1	0.0556
Industrial Metals	1	0.0434

Replication basket – AMLP+URA+GUNR (cor=0.98) basket lead at lag 1: 0.047

ticker	rep_wt	name	category	etf_signal
AMLP	0.139	Alerian MLP ETF	MLP / Energy Infrastr...	LONG
URA	0.095	Global X Uranium ETF	Materials	FLAT
GUNR	0.775	FlexShares Global Upstream Natural Re...	Materials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	GNR	0.0692	State Street SPDR S&P Global Natural Re...	Materials	LONG
10	GUNR	0.0689	FlexShares Global Upstream Natural Reso...	Materials	LONG
17	NANR	0.0668	State Street SPDR S&P North American Na...	Materials	LONG
16	IGE	0.0644	iShares North American Natural Resource...	Energy	LONG
9	MXI	0.0634	iShares Global Materials ETF	Materials	FLAT
12	PICK	0.0632	iShares MSCI Global Select Metals & Min...	Materials	FLAT
3	XME	0.0614	State Street SPDR S&P Metals & Mining E...	Materials	FLAT
13	SLX	0.0608	VanEck Steel ETF	Materials	LONG
5	MOO	0.0597	VanEck Agribusiness ETF	Consumer Staples	LONG
2	COPX	0.0589	Global X Copper Miners ETF	Materials	FLAT
8	MLPX	0.0565	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
14	NORW	0.0556	Global X MSCI Norway ETF	Norway Equity	FLAT
4	EMLP	0.0549	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	LONG
6	AMLP	0.0538	Alerian MLP ETF	MLP / Energy Infrastruct...	LONG
7	MLPA	0.0525	Global X MLP ETF	MLP / Energy Infrastruct...	LONG
1	URA	0.0467	Global X Uranium ETF	Materials	FLAT
15	DBB	0.0434	Invesco DB Base Metals Fund	Industrial Metals	FLAT

Cluster 13 – China Equity**Members:** 15 **Var explained:** 0.796 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-04-15 – filter **PASS** (252d at date = +24.3%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +0.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-04-15)

**Category breakdown**

category	n_members	abs_weight
China Equity	13	0.8765
Emerging Markets Equity	1	0.0644
Hong Kong Equity	1	0.0591

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.011

ticker	rep_wt	name	category	etf_signal
KWEB	0.562	KraneShares CSI China Internet ETF	China Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	CXSE	0.0740	WisdomTree China ex-State-Owned Enterpr...	China Equity	FLAT
9	GXC	0.0738	State Street SPDR S&P China ETF	China Equity	FLAT
1	MCHI	0.0732	iShares MSCI China ETF	China Equity	FLAT
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	FLAT
2	FXI	0.0711	iShares China Large-Cap ETF	China Equity	FLAT
7	CHIQ	0.0705	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0680	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0668	Xtrackers Harvest CSI 300 China A-Share...	China Equity	FLAT
8	CNYA	0.0660	iShares MSCI China A ETF	China Equity	FLAT
13	KGRN	0.0653	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0644	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
10	KBA	0.0596	KraneShares Boseri MSCI China A 50 Conn...	China Equity	LONG
12	EWI	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	LONG
14	CNXT	0.0588	VanEck ChiNext Innovators ETF	China Equity	FLAT
15	KURE	0.0582	KraneShares MSCI All China Health Care ...	China Equity	LONG

Cluster 14 – Financials

Members: 14 **Var explained:** 0.872 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +19.9%)
- Latest sell_cross: 2026-05-11
- Current 252d return: +11.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-11 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Financials	13	0.9382
Global Equity	1	0.0618

Replication basket – XLF (cor=0.96) basket lead at lag 1: 0.012

ticker	rep_wt	name	category	etf_signal
XLF	1.095	State Street Financial Select Sector ...	Financials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
10	VFH	0.0756	Vanguard Financials ETF	Financials	FLAT
11	FNCL	0.0756	Fidelity MSCI Financials Index ETF	Financials	LONG
7	IYF	0.0753	iShares U.S. Financial ETF	Financials	FLAT
5	FXO	0.0752	First Trust Financials AlphaDEX	Financials	LONG
8	IYG	0.0749	iShares U.S. Financial Services ETF	Financials	FLAT
1	XLF	0.0742	State Street Financial Select Sector SP...	Financials	LONG

9	KBWB	0.0733	Invesco KBW Bank ETF	Financials	FLAT
3	KBE	0.0712	State Street SPDR S&P Bank ETF	Financials	FLAT
14	KCE	0.0708	State Street SPDR S&P Capital Markets E...	Financials	LONG
6	IAT	0.0706	iShares U.S. Regional Banks ETF	Financials	LONG
13	IAI	0.0696	iShares U.S. Broker-Dealers & Securitie...	Financials	FLAT
2	KRE	0.0687	State Street SPDR S&P Regional Banking ...	Financials	FLAT
4	KIE	0.0630	State Street SPDR S&P Insurance ETF	Financials	LONG
12	PSP	0.0618	Invesco Global Listed Private Equity ETF	Global Equity	FLAT

Cluster 15 – Real Estate

Members: 14 **Var explained:** 0.961 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +14.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: -0.002

ticker	rep_wt	name	category	etf_signal
VNQ	0.971	Vanguard Real Estate ETF	Real Estate	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

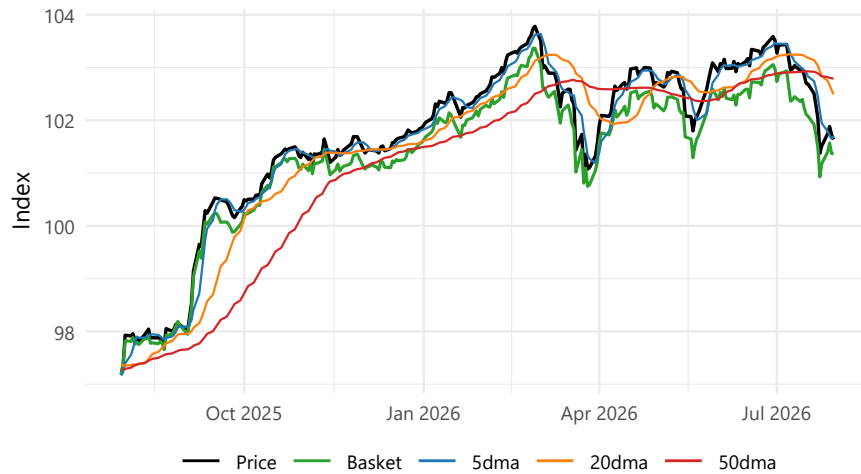
	ticker	PC_wt	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	FLAT
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	LONG
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	FLAT
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	LONG
10	USRT	0.0724	iShares Core U.S. REIT ETF	Real Estate	FLAT
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	FLAT
13	PSR	0.0722	Invesco Active U.S. Real Estate Fund	Real Estate	FLAT
1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	LONG
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	FLAT
7	REET	0.0715	iShares Global REIT ETF	Real Estate	FLAT
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	LONG
4	REZ	0.0697	iShares Residential and Multisector Rea...	Real Estate	FLAT
14	NURE	0.0688	Nuveen Short-Term REIT ETF	Real Estate	LONG
12	INDS	0.0683	Pacer Industrial Real Estate ETF	Real Estate	LONG

Cluster 16 – Municipal Bond

Members: 13 **Var explained:** 0.820 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-28 – filter **PASS** (252d at date = +5.9%)
- Latest sell_cross: 2026-07-10
- Current 252d return: +4.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-10 more recent than last effective buy 2026-05-28)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.180

ticker	rep_wt	name	category	etf_signal
MUB	0.904	iShares National Muni Bond ETF	Municipal Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	FLAT
6	CMF	0.0802	iShares California Muni Bond ETF	Municipal Bond	FLAT
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	FLAT
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	FLAT
1	FMB	0.0794	First Trust Managed Municipal ETF	Municipal Bond	FLAT
7	NYF	0.0794	iShares New York Muni Bond ETF	Municipal Bond	FLAT

5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	FLAT
4	MUNI	0.0770	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	FLAT
2	HYD	0.0747	VanEck High Yield Muni ETF	Municipal Bond	FLAT
3	HYMB	0.0721	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	FLAT
12	SUB	0.0694	iShares Short-Term National Muni Bond E...	Municipal Bond	FLAT
10	SHM	0.0667	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	FLAT

Cluster 17 – Gold

Members: 12 **Var explained:** 0.863 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-12 – filter **PASS** (252d at date = +75.1%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +26.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-05-12)



Category breakdown

category	n_members	abs_weight
----------	-----------	------------

Materials	5	0.4157
Gold	3	0.2502
Precious Metals	2	0.1723
Silver	2	0.1618

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.011

ticker	rep_wt	name	category	etf_signal
GLD	0.004	SPDR Gold Shares	Gold	FLAT
IAU	0.767	iShares Gold Trust Shares	Gold	FLAT
SLV	0.413	iShares Silver Trust	Silver	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	DBP	0.0865	Invesco DB Precious Metals Fund	Precious Metals	FLAT
7	GLTR	0.0858	abrdn Physical Precious Metals Basket S...	Precious Metals	FLAT
1	GDX	0.0840	VanEck Gold Miners ETF	Materials	FLAT
2	GDXJ	0.0840	VanEck Junior Gold Miners ETF	Materials	FLAT
3	GLD	0.0834	SPDR Gold Shares	Gold	FLAT
9	IAU	0.0834	iShares Gold Trust Shares	Gold	FLAT
10	SGOL	0.0833	abrdn Physical Gold Shares ETF	Gold	FLAT
12	GOAU	0.0831	US Global GO Gold and Precious Metal Mi...	Materials	FLAT
6	SIL	0.0828	Global X Silver Miners ETF	Materials	FLAT
5	SILJ	0.0818	Amplify Junior Silver Miners ETF	Materials	FLAT
4	SLV	0.0810	iShares Silver Trust	Silver	FLAT
8	SIVR	0.0808	abrdn Physical Silver Shares ETF	Silver	FLAT

Cluster 18 – Consumer Staples

Members: 11 **Var explained:** 0.742 **Signal:** LONG since 2026-06-26 (23 days, -0.3%)

Signal logic:

- Latest buy_cross: 2026-06-26 – filter **PASS** (252d at date = +9.9%)
- Latest sell_cross: 2026-05-29
- Current 252d return: +6.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-26 > last sell at 2026-05-29)



Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.5742
Utilities	4	0.3507
U.S. Small Cap Consumer Staples	1	0.0751

Replication basket – XLU+XLP (cor=0.99) basket lead at lag 1: -0.017

ticker	rep_wt	name	category	etf_signal
XLU	0.362	State Street Utilities Select Sector ...	Utilities	FLAT
XLP	0.588	State Street Consumer Staples Select ...	Consumer Staples	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	VDC	0.0984	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0984	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0976	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0951	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG

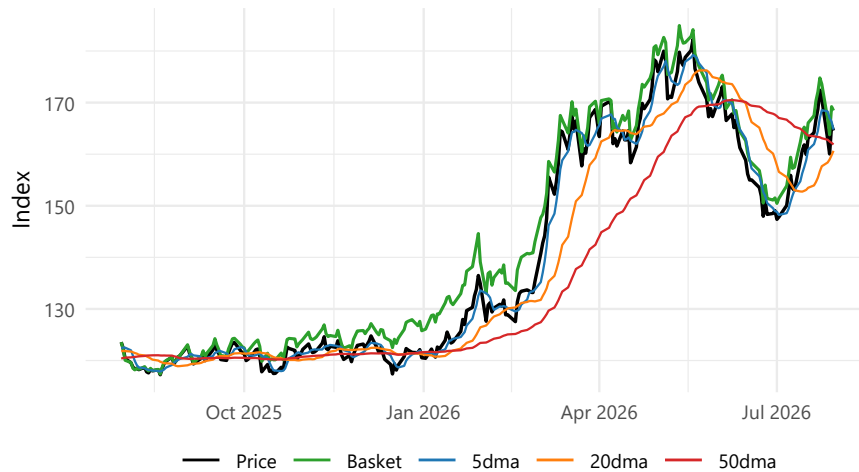
3	KXI	0.0947	iShares Global Consumer Staples ETF	Consumer Staples	LONG
11	PBJ	0.0900	Invesco Food & Beverage ETF	Consumer Staples	LONG
7	IDU	0.0887	iShares U.S. Utilities ETF	Utilities	LONG
9	FUTY	0.0875	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
8	VPU	0.0874	Vanguard Utilities ETF	Utilities	FLAT
2	XLU	0.0871	State Street Utilities Select Sector SP...	Utilities	FLAT
10	PSCC	0.0751	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	LONG

Cluster 19 – Broad Commodities

Members: 11 **Var explained:** 0.771 **Signal:** LONG since 2026-07-22 (6 days, -2.5%)

Signal logic:

- Latest buy_cross: 2026-07-22 – filter **PASS** (252d at date = +38.3%)
- Latest sell_cross: 2026-05-27
- Current 252d return: +32.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-22 > last sell at 2026-05-27)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6173
Oil	2	0.1980
Energy Commodities	1	0.1019
Agricultural Commodities	1	0.0497
Natural Gas	1	0.0331

Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.077

ticker	rep_wt	name	category	etf_signal
PDBC	1.124	Invesco Optimum Yield Diversified Com...	Broad Commodities	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	DBC	0.1056	Invesco DB Commodity Index Tracking Fund	Broad Commodities	LONG
6	GSG	0.1056	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	LONG
8	PDBC	0.1056	Invesco Optimum Yield Diversified Commo...	Broad Commodities	LONG
9	COMT	0.1050	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	LONG
10	DBE	0.1019	Invesco DB Energy Fund	Energy Commodities	LONG
1	USO	0.0996	United States Oil Fund	Oil	LONG
5	DBO	0.0984	Invesco DB Oil Fund	Oil	LONG
7	DJP	0.0984	iPath Bloomberg Commodity Index Total R...	Broad Commodities	LONG
4	FTGC	0.0971	First Trust Global Tactical Commodity S...	Broad Commodities	LONG
3	DBA	0.0497	Invesco DB Agriculture Fund	Agricultural Commodities	LONG
11	UNG	0.0331	United States Natural Gas Fund LP	Natural Gas	FLAT

Cluster 20 – India Equity

Members: 10 **Var explained:** 0.405 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +4.7%)
- Latest sell_cross: 2026-07-27
- Current 252d return: -12.4% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-27 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.4681
Malaysia Equity	1	0.1126
Indonesia Equity	1	0.1037
Philippines Equity	1	0.1013
Vietnam Equity	1	0.0750
Turkey Equity	1	0.0607
U.S. Dividend Equity	1	0.0547
Platinum	1	0.0238

Replication basket – NOBL+INDA+PPLT (cor=0.91) basket lead at lag 1: -0.005

ticker	rep_wt	name	category	etf_signal
NOBL	0.080	ProShares S&P 500 Dividend Aristocrat...	U.S. Dividend Equity	FLAT
INDA	0.660	iShares MSCI India ETF	India Equity	FLAT
PPLT	0.029	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	INDA	0.1574	iShares MSCI India ETF	India Equity	FLAT
2	INDY	0.1557	iShares India 50 ETF	India Equity	FLAT
3	EPI	0.1550	WisdomTree India Earnings Fund	India Equity	FLAT
8	EWM	0.1126	iShares MSCI Malaysia Index Fund	Malaysia Equity	LONG
6	EIDO	0.1037	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
7	EPHE	0.1013	iShares MSCI Philippines ETF	Philippines Equity	FLAT
10	VNM	0.0750	VanEck Vietnam ETF	Vietnam Equity	FLAT
9	TUR	0.0607	iShares MSCI Turkey ETF	Turkey Equity	FLAT
4	NOBL	0.0547	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
5	PPLT	0.0238	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Cluster 21 – Inflation-Protected Fixed Income

Members: 10 **Var explained:** 0.905 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-27 – filter **PASS** (252d at date = +4.8%)
- Latest sell_cross: 2026-07-02
- Current 252d return: +0.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-02 more recent than last effective buy 2026-05-27)



Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP+SCHP (cor=0.99) basket lead at lag 1: -0.014

ticker	rep_wt	name	category	etf_signal
VTIP	0.269	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT
SCHP	0.760	Schwab U.S. TIPS ETF	Inflation-Protected F...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	TDTF	0.1035	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	FLAT
6	TIPX	0.1034	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	FLAT
3	TIP	0.1030	iShares TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
8	SCHP	0.1023	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	FLAT
5	TDTT	0.1008	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.1000	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	FLAT

9	STIP	0.0998	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
2	STPZ	0.0996	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0990	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
1	LTPZ	0.0884	Pimco 15 Year U.S. TIPS Index Exchange-...	Inflation-Protected Fixe...	FLAT

Cluster 22 – Japan Equity

Members: 9 **Var explained:** 0.887 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +57.2%)
- Latest sell_cross: 2026-07-28
- Current 252d return: +35.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-28 more recent than last effective buy 2026-04-09)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7805

Asia-Pacific Equity	2	0.2195
---------------------	---	--------

Replication basket – EWJ (cor=0.97) basket lead at lag 1: -0.044

ticker	rep_wt	name	category	etf_signal
EWJ	0.907	iShares MSCI Japan Index Fund	Japan Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

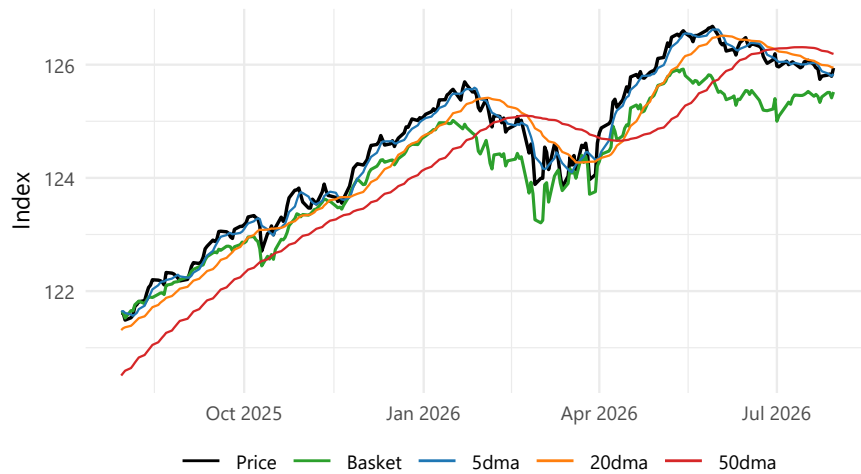
	ticker	PC_wt	name	category	etf_signal
1	EWJ	0.1142	iShares MSCI Japan Index Fund	Japan Equity	FLAT
8	BBJP	0.1142	JPMorgan BetaBuilders Japan ETF	Japan Equity	FLAT
6	DBJP	0.1118	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	FLAT
9	HEWJ	0.1118	iShares Currency Hedged MSCI Japan ETF	Japan Equity	FLAT
3	IPAC	0.1114	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	FLAT
5	FLJH	0.1110	Franklin FTSE Japan Hedged ETF	Japan Equity	FLAT
2	DXJ	0.1096	WisdomTree Japan Hedged Equity Fund	Japan Equity	FLAT
4	VPL	0.1082	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	FLAT
7	EWJV	0.1079	iShares MSCI Japan Value ETF	Japan Equity	LONG

Cluster 23 – Bank Loan / Floating Rate

Members: 9 **Var explained:** 0.556 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +3.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3621
Interest Rate Hedged Fixed Income	3	0.3244
Floating Rate Fixed Income	3	0.3134

Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.047

ticker	rep_wt	name	category	etf_signal
FLOT	0.291	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	LONG
BKLN	0.214	Invesco Senior Loan ETF	Bank Loan / Floating ...	FLAT
SRLN	0.302	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	LONG
FLRN	0.128	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	LONG
FLTR	0.035	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
2 SRLN	0.1250	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	LONG

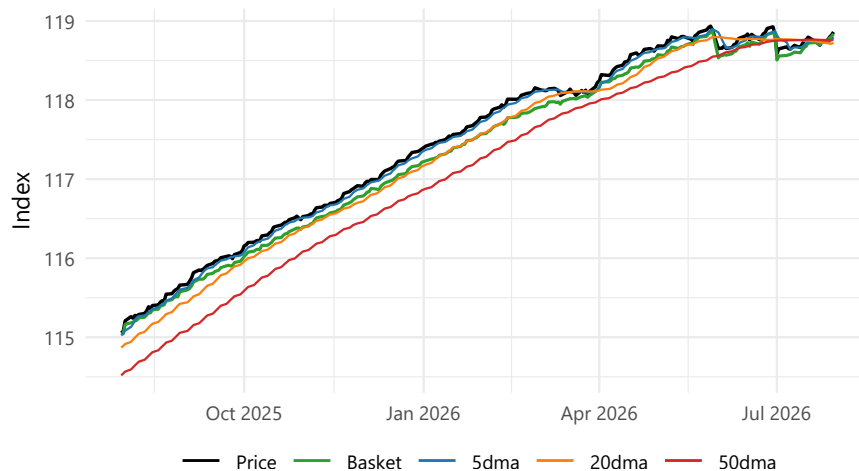
3	BKLN	0.1208	Invesco Senior Loan ETF	Bank Loan / Floating Rate	FLAT
1	FTSL	0.1163	First Trust Senior Loan Fund	Bank Loan / Floating Rate	FLAT
5	LQDH	0.1144	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	FLAT
7	FLOT	0.1084	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	LONG
6	IGBH	0.1056	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	FLAT
4	IGHG	0.1045	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	FLAT
8	FLRN	0.1039	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	LONG
9	FLTR	0.1011	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 24 – Cash / Ultra-Short Fixed Income

Members: 9 **Var explained:** 0.456 **Signal:** LONG since 2026-07-29 (1 days, 0.0%)

Signal logic:

- Latest buy_cross: 2026-07-29 – filter **PASS** (252d at date = +3.3%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +3.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-29 > last sell at 2026-07-07)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.85) basket lead at lag 1: 0.065

ticker	rep_wt	name	category	etf_signal
SHV	0.276	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	LONG
MINT	0.172	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	LONG
GBIL	0.122	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	LONG
ICSH	0.403	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	SHV	0.1309	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	LONG
6	ICSH	0.1296	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	LONG
5	GSY	0.1161	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	FLAT
2	MINT	0.1128	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	LONG
3	FTSM	0.1126	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	LONG
4	NEAR	0.1107	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT
8	BIL	0.1018	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	LONG
1	VNLA	0.0953	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	LONG
9	GBIL	0.0901	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	LONG

Cluster 25 – International Fixed Income

Members: 6 **Var explained:** 0.742 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-29 – filter **PASS** (252d at date = +0.6%)
- Latest sell_cross: 2026-07-23
- Current 252d return: -1.5% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-05-29)



Category breakdown

category	n_members	abs_weight
Currency	4	0.6411
International Fixed Income	2	0.3589

Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.026

ticker	rep_wt	name	category	etf_signal
BWX	0.752	State Street SPDR Bloomberg Internati...	International Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	IGOV	0.1800	iShares International Treasury Bond ETF	International Fixed Inco...	FLAT
1	BWX	0.1789	State Street SPDR Bloomberg Internation...	International Fixed Inco...	FLAT
4	FXE	0.1701	Invesco CurrencyShares Euro Currency Tr...	Currency	FLAT
5	FXF	0.1651	Invesco CurrencyShares Swiss Franc Trust	Currency	FLAT
3	FXB	0.1609	Invesco CurrencyShares British Pound St...	Currency	LONG
6	FXY	0.1449	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 26 – Leveraged / Inverse**Members:** 2 **Var explained:** 0.626 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-07-20 – filter **BLOCKED** (252d at date = -31.3%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell_cross: 2026-07-06
- Current 252d return: -29.5% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-06 more recent than last effective buy 2025-04-08)

**Category breakdown**

category	n_members	abs_weight
Currency	1	0.5
Leveraged / Inverse	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.024

ticker	rep_wt	name	category	etf_signal
SQQQ	0.506	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT

UUP	0.495	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT
-----	-------	---------------------------------------	----------	------

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
RWO	cluster_15	Real Estate	2026-06-25	24	3.6%	0.00	1.00
XLRE	cluster_15	Real Estate	2026-06-25	24	1.6%	0.00	1.00
IYR	cluster_15	Real Estate	2026-06-25	24	2.5%	0.00	1.00
FREL	cluster_15	Real Estate	2026-06-25	24	2.4%	0.00	1.00
INDS	cluster_15	Real Estate	2026-06-08	36	6.7%	0.00	1.00
NURE	cluster_15	Real Estate	2026-04-13	75	13.3%	0.06	0.94
SCHD	cluster_3	U.S. Large Cap Value	2026-07-07	17	2.7%	0.00	1.00
FDL	cluster_3	U.S. Large Cap Value	2026-07-16	10	0.4%	0.00	1.00
HDV	cluster_3	U.S. Large Cap Value	2026-06-25	24	4.3%	0.00	1.00
FTA	cluster_3	U.S. Large Cap Value	2026-05-21	47	7.7%	0.00	1.00
PWV	cluster_3	U.S. Large Cap Value	2026-04-02	81	14.5%	0.00	1.00
DIV	cluster_3	U.S. Large Cap Value	2026-06-29	22	2.7%	0.00	1.00
FVD	cluster_3	U.S. Large Cap Value	2026-06-10	34	5.2%	0.00	1.00
PKW	cluster_3	U.S. Large Cap Value	2026-05-20	48	7.9%	0.00	1.00
LGLV	cluster_3	U.S. Large Cap Value	2026-06-26	23	2.4%	0.00	1.00
ACWV	cluster_3	U.S. Large Cap Value	2026-07-06	18	2.2%	0.00	1.00
DGRO	cluster_3	U.S. Large Cap Value	2026-04-13	75	8.2%	0.00	1.00
FTCS	cluster_3	U.S. Large Cap Value	2026-06-26	23	5.7%	0.00	1.00
SDY	cluster_3	U.S. Large Cap Value	2026-05-20	48	4.9%	0.00	1.00
DTD	cluster_3	U.S. Large Cap Value	2026-04-13	75	6.9%	0.00	1.00
FNDX	cluster_3	U.S. Large Cap Value	2026-04-09	77	10.1%	0.00	1.00
SPHD	cluster_3	U.S. Large Cap Value	2026-06-08	36	5.1%	0.01	0.99
VONV	cluster_3	U.S. Large Cap Value	2026-04-10	76	13.0%	0.01	0.99
IWD	cluster_3	U.S. Large Cap Value	2026-04-10	76	13.0%	0.01	0.99
OUSA	cluster_3	U.S. Large Cap Value	2026-07-01	20	3.5%	0.01	0.99
LVHD	cluster_3	U.S. Large Cap Value	2026-06-10	34	3.1%	0.01	0.99
DLN	cluster_3	U.S. Large Cap Value	2026-04-13	75	7.0%	0.02	0.98
VTV	cluster_3	U.S. Large Cap Value	2026-04-13	75	8.8%	0.02	0.98
RWL	cluster_3	U.S. Large Cap Value	2026-04-10	76	10.8%	0.02	0.98
IWX	cluster_3	U.S. Large Cap Value	2026-04-10	76	13.9%	0.02	0.98
SPLV	cluster_3	U.S. Large Cap Value	2026-06-11	33	3.4%	0.03	0.97
VIG	cluster_3	U.S. Large Cap Value	2026-04-14	74	7.2%	0.04	0.96
AIVL	cluster_3	U.S. Large Cap Value	2026-04-10	76	9.7%	0.05	0.95
RSP	cluster_3	U.S. Large Cap Value	2026-04-14	74	7.9%	0.09	0.91
PRF	cluster_3	U.S. Large Cap Value	2026-04-09	77	10.5%	0.18	0.82
VLUE	cluster_3	U.S. Large Cap Value	2026-07-27	3	-1.0%	0.58	-0.57
FEX	cluster_3	U.S. Large Cap Value	2026-04-09	77	7.5%	0.72	-0.67

JVAL	cluster_3	U.S. Large Cap Value	2026-07-30	0	0.0%	0.91	-0.90
BBAX	cluster_8	Asia-Pacific Equity	2026-07-14	12	4.2%	0.00	1.00
EPP	cluster_8	Asia-Pacific Equity	2026-07-15	11	3.2%	0.00	1.00
EFAV	cluster_8	Asia-Pacific Equity	2026-07-16	10	4.1%	0.00	1.00
EWA	cluster_8	Asia-Pacific Equity	2026-07-20	8	4.6%	0.00	1.00
PID	cluster_8	Asia-Pacific Equity	2026-07-16	10	1.8%	0.00	1.00
IDLV	cluster_8	Asia-Pacific Equity	2026-07-10	14	4.3%	0.00	1.00
DWX	cluster_8	Asia-Pacific Equity	2026-07-09	15	4.1%	0.00	1.00
RWX	cluster_8	Asia-Pacific Equity	2026-07-16	10	3.3%	0.00	1.00
EWC	cluster_8	Asia-Pacific Equity	2026-07-10	14	1.9%	0.00	1.00
ENZL	cluster_8	Asia-Pacific Equity	2026-07-07	17	4.0%	0.00	1.00
EWS	cluster_8	Asia-Pacific Equity	2026-06-17	29	9.5%	0.00	1.00
ILF	cluster_8	Asia-Pacific Equity	2026-07-22	6	-0.1%	0.01	0.99
VNQI	cluster_8	Asia-Pacific Equity	2026-07-20	8	2.3%	0.01	0.99
EWZ	cluster_8	Asia-Pacific Equity	2026-07-16	10	3.4%	0.05	0.95
NFRA	cluster_8	Asia-Pacific Equity	2026-07-16	10	-0.0%	0.05	0.91
EWL	cluster_8	Asia-Pacific Equity	2026-07-28	2	1.1%	0.73	0.08
IGF	cluster_8	Asia-Pacific Equity	2026-07-09	15	0.4%	0.83	-0.11
DBO	cluster_19	Broad Commodities	2026-07-21	7	-0.9%	0.00	1.00
GSG	cluster_19	Broad Commodities	2026-07-21	7	-0.7%	0.00	1.00
DBE	cluster_19	Broad Commodities	2026-07-20	8	1.0%	0.00	1.00
USO	cluster_19	Broad Commodities	2026-07-22	6	-3.2%	0.00	1.00
PDBC	cluster_19	Broad Commodities	2026-07-21	7	-0.8%	0.00	1.00
DBC	cluster_19	Broad Commodities	2026-07-21	7	-0.8%	0.00	1.00
DJP	cluster_19	Broad Commodities	2026-07-22	6	-2.2%	0.01	0.99
COMT	cluster_19	Broad Commodities	2026-07-23	5	-1.6%	0.01	0.99
FTGC	cluster_19	Broad Commodities	2026-07-17	9	0.3%	0.02	0.98
DBA	cluster_19	Broad Commodities	2026-07-09	15	-0.8%	0.61	0.39
IYK	cluster_18	Consumer Staples	2026-06-09	35	3.9%	0.00	1.00
XLP	cluster_18	Consumer Staples	2026-07-27	3	0.1%	0.00	1.00
PSCC	cluster_18	Consumer Staples	2026-06-10	34	8.3%	0.01	0.99
FSTA	cluster_18	Consumer Staples	2026-07-28	2	-1.8%	0.01	0.99
VDC	cluster_18	Consumer Staples	2026-07-28	2	-1.7%	0.01	0.99
KXI	cluster_18	Consumer Staples	2026-07-28	2	-0.7%	0.03	0.97
PBJ	cluster_18	Consumer Staples	2026-07-13	13	-0.3%	0.17	0.83
IDU	cluster_18	Consumer Staples	2026-06-25	24	-2.4%	0.88	0.05
FXO	cluster_14	Financials	2026-06-05	37	10.6%	0.00	1.00
XLF	cluster_14	Financials	2026-04-13	75	10.3%	0.04	0.96
KCE	cluster_14	Financials	2026-07-07	17	2.1%	0.05	0.95
FNCL	cluster_14	Financials	2026-04-13	75	10.3%	0.05	0.95
KIE	cluster_14	Financials	2026-06-10	34	10.9%	0.11	0.89
IAT	cluster_14	Financials	2026-05-20	48	11.6%	0.80	0.20
GNR	cluster_12	Materials	2026-07-27	3	2.7%	0.00	1.00
GUNR	cluster_12	Materials	2026-07-27	3	2.3%	0.00	1.00
NANR	cluster_12	Materials	2026-07-24	4	0.2%	0.00	1.00

IGE	cluster_12	Materials	2026-07-23	5	-0.4%	0.00	1.00
MOO	cluster_12	Materials	2026-07-07	17	3.2%	0.00	1.00
AML	cluster_12	Materials	2026-07-10	14	4.0%	0.00	1.00
MLPA	cluster_12	Materials	2026-07-10	14	4.0%	0.00	1.00
SLX	cluster_12	Materials	2026-07-30	0	0.0%	0.00	0.98
EMLP	cluster_12	Materials	2026-07-08	16	-0.6%	0.46	0.53
MLPX	cluster_12	Materials	2026-07-09	15	-0.4%	0.79	0.02
EWM	cluster_20	India Equity	2026-07-30	0	0.0%	0.02	0.98
BIL	cluster_24	Cash / Ultra-Short Fixe...	2026-07-16	10	0.1%	0.01	0.97
SHV	cluster_24	Cash / Ultra-Short Fixe...	2026-07-20	8	0.1%	0.02	0.96
GBIL	cluster_24	Cash / Ultra-Short Fixe...	2026-07-20	8	0.1%	0.01	0.94
MINT	cluster_24	Cash / Ultra-Short Fixe...	2026-07-20	8	0.1%	0.03	0.91
FTSM	cluster_24	Cash / Ultra-Short Fixe...	2026-07-27	3	0.1%	0.02	0.89
ICSH	cluster_24	Cash / Ultra-Short Fixe...	2026-07-28	2	0.0%	0.11	0.59
VNLA	cluster_24	Cash / Ultra-Short Fixe...	2026-07-30	0	0.0%	0.09	0.54
EWI	cluster_13	China Equity	2026-07-23	5	3.8%	0.00	1.00
KURE	cluster_13	China Equity	2026-07-06	18	0.8%	0.48	0.52
KBA	cluster_13	China Equity	2026-07-27	3	-1.8%	0.50	-0.38
HDEF	cluster_4	Developed ex-US Equity	2026-07-09	15	4.8%	0.00	1.00
IDV	cluster_4	Developed ex-US Equity	2026-07-28	2	2.1%	0.00	1.00
VYMI	cluster_4	Developed ex-US Equity	2026-07-08	16	4.5%	0.00	1.00
RODM	cluster_4	Developed ex-US Equity	2026-07-27	3	1.4%	0.00	1.00
JPIN	cluster_4	Developed ex-US Equity	2026-07-27	3	2.6%	0.00	1.00
EPOL	cluster_4	Developed ex-US Equity	2026-07-08	16	7.5%	0.00	1.00
IPKW	cluster_4	Developed ex-US Equity	2026-07-10	14	4.7%	0.00	1.00
EFV	cluster_4	Developed ex-US Equity	2026-07-08	16	4.8%	0.00	1.00
EWU	cluster_4	Developed ex-US Equity	2026-07-16	10	3.6%	0.00	1.00
VIGI	cluster_4	Developed ex-US Equity	2026-06-30	21	5.0%	0.00	1.00
QEFA	cluster_4	Developed ex-US Equity	2026-07-06	18	2.5%	0.00	1.00
EWD	cluster_4	Developed ex-US Equity	2026-07-30	0	0.0%	0.00	0.99
IVLU	cluster_4	Developed ex-US Equity	2026-07-27	3	2.3%	0.00	0.99
EUFN	cluster_4	Developed ex-US Equity	2026-06-12	32	7.3%	0.02	0.98
GSIE	cluster_4	Developed ex-US Equity	2026-07-02	19	3.0%	0.02	0.97
TLTD	cluster_4	Developed ex-US Equity	2026-07-30	0	0.0%	0.01	0.96
INTF	cluster_4	Developed ex-US Equity	2026-07-27	3	2.0%	0.04	0.94
DWM	cluster_4	Developed ex-US Equity	2026-07-30	0	0.0%	0.04	0.91
VGK	cluster_4	Developed ex-US Equity	2026-07-01	20	3.7%	0.08	0.87
IQLT	cluster_4	Developed ex-US Equity	2026-07-01	20	3.2%	0.13	0.86
IEV	cluster_4	Developed ex-US Equity	2026-07-27	3	2.4%	0.15	0.79
IEUR	cluster_4	Developed ex-US Equity	2026-07-30	0	0.0%	0.13	0.77
EWQ	cluster_4	Developed ex-US Equity	2026-07-30	0	0.0%	0.04	0.74
ESGD	cluster_4	Developed ex-US Equity	2026-07-30	0	0.0%	0.23	0.63
EWK	cluster_4	Developed ex-US Equity	2026-07-29	1	-0.3%	0.02	0.53
EFA	cluster_4	Developed ex-US Equity	2026-07-30	0	0.0%	0.26	0.50
IEFA	cluster_4	Developed ex-US Equity	2026-07-30	0	0.0%	0.26	0.42

FEZ	cluster_4	Developed ex-US Equity	2026-07-30	0	0.0%	0.28	0.34
IQDF	cluster_4	Developed ex-US Equity	2026-07-15	11	-0.0%	0.68	-0.64
HEZU	cluster_4	Developed ex-US Equity	2026-07-30	0	0.0%	0.75	-0.70
PPH	cluster_11	Healthcare	2026-05-21	47	7.5%	0.04	0.96
XLV	cluster_11	Healthcare	2026-05-21	47	10.4%	0.10	0.90
IYH	cluster_11	Healthcare	2026-05-20	48	11.2%	0.21	0.79
FHLC	cluster_11	Healthcare	2026-05-19	49	12.2%	0.24	0.76
VHT	cluster_11	Healthcare	2026-05-19	49	12.1%	0.26	0.74
IHF	cluster_11	Healthcare	2026-05-08	56	13.6%	0.86	0.14
PTH	cluster_11	Healthcare	2026-06-17	29	12.3%	1.00	-0.01
XMLV	cluster_1	U.S. Mid Cap Equity	2026-06-25	24	3.6%	0.00	1.00
RSPM	cluster_1	U.S. Mid Cap Equity	2026-07-27	3	0.6%	0.00	1.00
REGL	cluster_1	U.S. Mid Cap Equity	2026-06-12	32	3.6%	0.00	1.00
XLB	cluster_1	U.S. Mid Cap Equity	2026-07-28	2	-1.3%	0.01	0.97
RSPD	cluster_1	U.S. Mid Cap Equity	2026-07-28	2	-0.8%	0.04	0.95
XRT	cluster_1	U.S. Mid Cap Equity	2026-05-28	43	5.8%	0.05	0.95
SDVY	cluster_1	U.S. Mid Cap Equity	2026-06-08	36	5.7%	0.06	0.94
RTH	cluster_1	U.S. Mid Cap Equity	2026-07-29	1	-0.8%	0.08	0.90
SMMV	cluster_1	U.S. Mid Cap Equity	2026-06-25	24	4.2%	0.16	0.84
IJS	cluster_1	U.S. Mid Cap Equity	2026-04-10	76	10.5%	0.59	0.39
ISMD	cluster_1	U.S. Mid Cap Equity	2026-04-09	77	16.0%	0.97	-0.09
PAVE	cluster_1	U.S. Mid Cap Equity	2026-07-27	3	-2.4%	0.61	-0.53
ITB	cluster_1	U.S. Mid Cap Equity	2026-07-28	2	-4.8%	0.79	-0.57
FTXR	cluster_1	U.S. Mid Cap Equity	2026-04-17	71	7.1%	0.89	-0.66
VIS	cluster_1	U.S. Mid Cap Equity	2026-07-23	5	-2.2%	0.93	-0.85
FIDU	cluster_1	U.S. Mid Cap Equity	2026-07-23	5	-2.4%	0.93	-0.85
FXB	cluster_25	International Fixed Inc...	2026-07-15	11	-0.4%	0.68	0.03
SKYY	cluster_10	Technology	2026-07-30	0	0.0%	0.42	0.18
EWJV	cluster_22	Japan Equity	2026-07-06	18	2.0%	0.02	0.98
SRLN	cluster_23	Bank Loan / Floating Ra...	2026-07-30	0	0.0%	0.04	0.95
FLOT	cluster_23	Bank Loan / Floating Ra...	2026-07-10	14	0.2%	0.01	0.95
FLRN	cluster_23	Bank Loan / Floating Ra...	2026-07-13	13	0.2%	0.04	0.87
IBDR	cluster_5	Aggregate Fixed Income	2026-07-21	7	0.2%	0.01	0.97
DVYE	cluster_7	Emerging Markets Equity	2026-07-24	4	1.5%	0.00	1.00
SDIV	cluster_7	Emerging Markets Equity	2026-07-21	7	0.5%	0.07	0.93
FNDE	cluster_7	Emerging Markets Equity	2026-07-30	0	0.0%	0.02	0.93
AIA	cluster_7	Emerging Markets Equity	2026-07-10	14	-5.2%	1.00	-1.00
ITA	cluster_2	U.S. Large Cap Equity	2026-07-24	4	-0.8%	0.30	0.55
PPA	cluster_2	U.S. Large Cap Equity	2026-07-27	3	-2.6%	0.33	0.31
PTF	cluster_6	U.S. Large Cap Growth	2026-04-07	79	4.3%	1.00	-1.00
QTUM	cluster_6	U.S. Large Cap Growth	2026-04-10	76	20.2%	1.00	-1.00

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
EWI	cluster_13	China Equity	1.00	1.00
GNR	cluster_12	Materials	1.00	1.00
RDIV	cluster_3	U.S. Large Cap Value	1.00	1.00
BBAX	cluster_8	Asia-Pacific Equity	1.00	1.00
HDEF	cluster_4	Developed ex-US Equity	1.00	1.00
PSR	cluster_15	Real Estate	1.00	1.00
DVYE	cluster_7	Emerging Markets Equity	1.00	1.00
RFV	cluster_1	U.S. Mid Cap Equity	1.00	1.00
PSP	cluster_14	Financials	1.00	1.00
DBO	cluster_19	Broad Commodities	1.00	1.00
IYK	cluster_18	Consumer Staples	1.00	1.00
EWJV	cluster_22	Japan Equity	1.00	0.98
IBDR	cluster_5	Aggregate Fixed Income	0.98	0.97
BIL	cluster_24	Cash / Ultra-Short Fixed ...	0.98	0.97
PPH	cluster_11	Healthcare	1.00	0.96
SRLN	cluster_23	Bank Loan / Floating Rate	0.99	0.95
QDF	cluster_2	U.S. Large Cap Equity	1.00	0.85
GDX	cluster_17	Gold	0.04	-0.08
VTIP	cluster_21	Inflation-Protected Fixed...	0.00	-0.47
HYEM	cluster_9	High Yield Bond	0.00	-0.74

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
HYD	cluster_16	Municipal Bond	1.00	-1.00
PCY	cluster_9	High Yield Bond	1.00	-1.00
EWX	cluster_7	Emerging Markets Equity	1.00	-1.00
ARKK	cluster_10	Technology	1.00	-1.00
PTNQ	cluster_6	U.S. Large Cap Growth	1.00	-1.00
IWO	cluster_1	U.S. Mid Cap Equity	1.00	-1.00

FPX	cluster_2	U.S. Large Cap Equity	1.00	-1.00
CQQQ	cluster_13	China Equity	1.00	-1.00
VNM	cluster_20	India Equity	1.00	-1.00
AGGY	cluster_5	Aggregate Fixed Income	1.00	-1.00
IGBH	cluster_23	Bank Loan / Floating Rate	1.00	-1.00
UNG	cluster_19	Broad Commodities	1.00	-1.00
VPL	cluster_22	Japan Equity	1.00	-1.00
EWN	cluster_4	Developed ex-US Equity	1.00	-1.00
LTPZ	cluster_21	Inflation-Protected Fixed...	0.99	-0.99
URA	cluster_12	Materials	0.97	-0.97
ARKG	cluster_11	Healthcare	1.00	-0.95
FXF	cluster_25	International Fixed Income	0.92	-0.92
EMLC	cluster_8	Asia-Pacific Equity	0.93	-0.90
JVAL	cluster_3	U.S. Large Cap Value	0.91	-0.90

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_19	11	FLAT	0.481	0.076	0.00017	
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_20	10	FLAT	0.305	0.095	0.00022	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.741	0.038	0.00008	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.634	0.213	0.00036	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_25	6	FLAT	0.730	0.095	0.00023	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_19	11	LONG	0.599	0.168	0.00039	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.743	0.126	0.00031	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.743	0.153	0.00040	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.699	0.191	0.00046	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.714	0.178	0.00044	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.679	0.179	0.00047	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_20	10	FLAT	0.501	0.218	0.00056	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.869	0.021	0.00001	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_5	58	FLAT	0.847	0.127	0.00036	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.869	0.075	0.00021	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_5	58	FLAT	0.876	0.162	0.00031	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_25	6	FLAT	0.821	0.204	0.00049	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_5	58	FLAT	0.832	0.233	0.00047	
NOBL	ProShares S&P 500 Dividend Aris...	U.S. Dividend Equity	cluster_20	10	FLAT	0.448	0.272	0.00075	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	13	FLAT	0.877	0.247	0.00046	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_20	10	FLAT	0.441	0.325	0.00083	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_13	15	LONG	0.791	0.268	0.00071	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	13	FLAT	0.877	0.263	0.00055	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_12	17	FLAT	0.746	0.359	0.00085	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_20	10	FLAT	0.495	0.348	0.00102	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_23	9	FLAT	0.865	0.323	0.00071	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_13	15	LONG	0.857	0.255	0.00073	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_20	10	FLAT	0.556	0.404	0.00107	
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected F...	cluster_21	10	FLAT	0.918	0.226	0.00057	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_13	15	FLAT	0.903	0.237	0.00064	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20